LAB2 PROGRAMS

1)

```
#include <stdio.h>

struct Process {
    int pid;
    int arrival;
    int burst;
    int completion;
    int turnaround;
    int waiting;
};

int main() {
    int n, i, j;
    float avg_waiting = 0, avg_turnaround = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process proc[n];

    printf("Enter ProcessID, ArrivalTime, BurstTime for %d processes:\n", n);
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i+1);
        scanf("%d %d %d", &proc[i].pid, &proc[i].arrival, &proc[i].burst);
    }
```

```
for(i = 0; i < n-1; i++) {
    for(j = 0; j < n-i-1; j++) {
        if(proc[j].arrival > proc[j+1].arrival) {
            struct Process temp = proc[j];
            proc[j] = proc[j+1];
            proc[j+1] = temp;
        }
    }
}

int current_time = 0;

for(i = 0; i < n; i++) {
    if(current_time < proc[i].arrival) {
        current_time = proc[i].arrival;
    }

    proc[i].completion = current_time + proc[i].burst;
    proc[i].turnaround = proc[i].completion - proc[i].arrival;
    proc[i].waiting = proc[i].turnaround - proc[i].burst;
    current_time = proc[i].completion;

    avg_waiting += proc[i].waiting;
    avg_turnaround += proc[i].turnaround;
}

printf("\nProcess Scheduling - FCFS Algorithm\n");
printf("====================================\n");
printf("PID\tArrival\tBurst\tCompletion\tTurnaround\tWaiting\n");
printf("-------------------------------------------------------\n");
```

```
    for(i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t%d\t\t%d\t\t%d\n",
            proc[i].pid, proc[i].arrival, proc[i].burst,
            proc[i].completion, proc[i].turnaround, proc[i].waiting);
    }

    printf("\nAverage Waiting Time: %.2f\n", avg_waiting/n);
    printf("Average Turnaround Time: %.2f\n", avg_turnaround/n);

    return 0;
}
```

```
C:\Users\Admin\Desktop\1wa

Enter number of processes: 4
Enter ProcessID, ArrivalTime, BurstTime for 4 processes:
Process 1: 1 0 8
Process 2: 2 3 9
Process 3: 3 5 7
Process 4: 4 9 4

Process Scheduling - FCFS Algorithm
====================================
PID     Arrival Burst   Completion      Turnaround      Waiting
-------------------------------------------------------
1       0       8       8               8               0
2       3       9       17              14              5
3       5       7       24              19              12
4       9       4       28              19              15

Average Waiting Time: 8.00
Average Turnaround Time: 15.00

Process returned 0 (0x0)   execution time : 95.283 s
Press any key to continue.
```

2)

```
#include <stdio.h>

int main() {
```

```
int n, i;
int pid[10], arrival[10], burst[10], ct[10], tat[10], wt[10];
int done[10] = {0};
float avg_wt = 0, avg_tat = 0;

printf("Enter number of processes: ");
scanf("%d", &n);

for(i = 0; i < n; i++) {
   printf("P%d: PID AT BT = ", i+1);
   scanf("%d%d%d", &pid[i], &arrival[i], &burst[i]);
}

int time = 0;
while(1) {
   int min = 999, pos = -1;

   for(i = 0; i < n; i++) {
      if(!done[i] && arrival[i] <= time && burst[i] < min) {
         min = burst[i];
         pos = i;
      }
   }

   if(pos == -1) {
      time++;
      int all_done = 1;
      for(i = 0; i < n; i++) if(!done[i]) all_done = 0;
      if(all_done) break;
   } else {
      ct[pos] = time + burst[pos];
```

```
            tat[pos] = ct[pos] - arrival[pos];
            wt[pos] = tat[pos] - burst[pos];
            done[pos] = 1;
            time = ct[pos];
        }
    }

    printf("\nPID AT BT CT TAT WT\n");
    for(i = 0; i < n; i++) {
        printf("%d  %d %d %d %d %d\n", pid[i], arrival[i], burst[i], ct[i], tat[i], wt[i]);
        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    printf("\nAvg WT: %.2f\n", avg_wt/n);
    printf("Avg TAT: %.2f\n", avg_tat/n);

    return 0;
}
```

```
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Enter number of processes: 4
P1: PID AT BT = 1 0 6
P2: PID AT BT = 2 2 4
P3: PID AT BT = 3 4 2
P4: PID AT BT = 4 6 8

PID AT BT CT TAT WT
1  0 6 6 6 0
2  2 4 12 10 6
3  4 2 8 4 2
4  6 8 20 14 6

Avg WT: 3.50
Avg TAT: 8.50

Process returned 0 (0x0)   execution time : 102.129 s
Press any key to continue.
```

3)

```
#include <stdio.h>


int main() {
    int n, i;
    int pid[10], arrival[10], burst[10], remaining[10];
    int ct[10], tat[10], wt[10];
    int done[10] = {0};
    float avg_wt = 0, avg_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("P%d: PID AT BT = ", i+1);
        scanf("%d%d%d", &pid[i], &arrival[i], &burst[i]);
```

```
    remaining[i] = burst[i];
}

int time = 0;
while(1) {
    int all_done = 1;
    for(i = 0; i < n; i++) {
        if(!done[i]) all_done = 0;
    }
    if(all_done) break;

    int min = 999, pos = -1;
    for(i = 0; i < n; i++) {
        if(!done[i] && arrival[i] <= time && remaining[i] < min) {
            min = remaining[i];
            pos = i;
        }
    }

    if(pos == -1) {
        time++;
    } else {
        remaining[pos]--;
        time++;
        if(remaining[pos] == 0) {
            ct[pos] = time;
            tat[pos] = ct[pos] - arrival[pos];
            wt[pos] = tat[pos] - burst[pos];
            done[pos] = 1;
        }
    }
```

```
    }

    printf("\nPID AT BT CT TAT WT\n");
    for(i = 0; i < n; i++) {
        printf("%d  %d %d %d %d %d\n", pid[i], arrival[i], burst[i], ct[i], tat[i], wt[i]);
        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    printf("\nAvg WT: %.2f\n", avg_wt/n);
    printf("Avg TAT: %.2f\n", avg_tat/n);

    return 0;
}
```

```
C:\Users\Admin\Desktop\1wa

Enter number of processes: 4
P1: PID AT BT = 1 0 6
P2: PID AT BT = 2 2 4
P3: PID AT BT = 3 4 2
P4: PID AT BT = 4 6 3

PID AT BT CT TAT WT
1  0 6 6 6 0
2  2 4 15 13 9
3  4 2 8 4 2
4  6 3 11 5 2

Avg WT: 3.25
Avg TAT: 7.00

Process returned 0 (0x0)   execution time : 65.546 s
Press any key to continue.
```